

1. The required IP address 172.24.3.164; and the port number is 59316.

The image shows a Wireshark packet capture of an HTTP POST request. The packet list at the top shows two packets: packet 147 is an HTTP POST request from 172.24.3.164 to 128.119.245.12 on port 80, and packet 158 is the corresponding HTTP 200 OK response. The packet details pane for packet 147 shows the following information:

- Frame 147: 1190 bytes on wire (9520 bits), 1190 bytes captured (9520 bits) on interface en0, id 0
- Ethernet II, Src: Apple_d5:84:fc (6c:7e:67:ad:50:4c), Dst: Fortinet_09:00:20 (00:09:0f:09:00:20)
- Internet Protocol Version 4, Src: 172.24.3.164, Dst: 128.119.245.12
- Transmission Control Protocol, Src Port: 59316, Dst Port: 80, Seq: 151801, Ack: 1, Len: 1136
- Source Port: 59316
- Destination Port: 80
- [Stream index: 0]
- [Conversation completeness: Incomplete, DATA (15)]
- [TCP Segment Len: 1136]
- Sequence Number: 151801 (relative sequence number)
- Sequence Number (raw): 3994540705
- [Next Sequence Number: 152937 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number)
- Acknowledgment number (raw): 589259183
- 0101 = Header Length: 20 bytes (5)
- Flags: 0x018 (PSH, ACK)
- 000. = Reserved: Not set
- ...0 = Accurate ECN: Not set
-0... = Congestion Window Reduced: Not set
-0... = ECN-Echo: Not set
-0... = Urgent: Not set
-1... = Acknowledgment: Set
-1... = Push: Set
-0... = Reset: Not set
-0... = Syn: Not set
-0... = Fin: Not set
- [TCP Flags:AP...]
- Window: 4096
- [Calculated window size: 262144]
- [Window size scaling factor: 64]
- Checksum: 0x32b9 (unverified)
- [Checksum Status: Unverified]

The packet bytes pane shows the raw data of the HTTP request, which is a POST request to /wireshark-labs/lab3-1-reply.htm. The data is shown in hexadecimal and ASCII format.

2. The required IP address 128.119.245.12; and the port number is 80.

The image shows a Wireshark packet capture of an HTTP transaction. The top pane displays a list of packets, with packet 158 selected. The middle pane shows the details of the selected packet, and the bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
147	1.842363	172.24.3.164	128.119.245.12	HTTP	1190	POST /wireshark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
158	2.259274	128.119.245.12	172.24.3.164	HTTP	831	HTTP/1.1 200 OK (text/html)

Packet Details:

- Frame 158: 831 bytes on wire (6648 bits), 831 bytes captured (6648 bits) on interface en0, id 0
- Ethernet II, Src: Fortinet_09:00:20 (00:09:0f:09:00:20), Dst: Apple_d5:04:fc (6c:7e:67:d5:04:fc)
- Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.24.3.164
- Transmission Control Protocol, Src Port: 80, Dst Port: 59316, Seq: 1, Ack: 152937, Len: 777
 - Source Port: 80
 - Destination Port: 59316
 - [Stream index: 0]
 - [Conversation completeness: Incomplete, DATA (15)]
 - [TCP Segment Len: 777]
 - Sequence Number: 1 (relative sequence number)
 - Sequence Number (raw): 580259183
 - [Next Sequence Number: 778 (relative sequence number)]
 - Acknowledgment Number: 152937 (relative ack number)
 - Acknowledgment number (raw): 3994541841
 - 0101 = Header Length: 20 bytes (5)
 - Flags: 0x018 (PSH, ACK)
 - 000. = Reserved: Not set
 - ...0 = Accurate ECN: Not set
 -0. = Congestion Window Reduced: Not set
 -0. = ECN-Echo: Not set
 -0. = Urgent: Not set
 -1. = Acknowledgment: Set
 -1. = Push: Set
 -0. = Reset: Not set
 -0. = Syn: Not set
 -0. = Fin: Not set
 - [TCP Flags:AP...]
 - Window: 1432
 - [Calculated window size: 183296]
 - [Window size scaling factor: 128]
 - Checksum: 0x9954 [unverified]
 - [Checksum Status: Unverified]

Raw Data:

0020 03 a4 00 50 e7 b4 23 1f 61 af ee 17 df 11 50 18 ...P-#- a-....P
0030 05 98 99 54 00 00 48 54 54 50 2f 31 2e 31 20 32 ...T-..HT TP/1.1 2
0040 30 30 20 4f 4b 0d 0a 44 61 74 65 3a 20 54 75 65 00 OK Date: Tue
0050 2c 20 31 30 20 4f 63 74 20 32 30 32 33 20 30 32 , 10 Oct 2023 02
0060 3a 30 35 3a 35 34 20 47 48 54 0d 0a 53 65 72 76 105:54 G MT. Serv
0070 65 72 3a 20 41 70 61 63 68 65 2f 32 2e 34 2e 36 err: Apac he/2.4.6
0080 20 28 43 65 66 74 4f 53 29 20 4f 70 65 66 53 53 (CentOS) OpenSS
0090 4c 2f 31 2e 30 2e 32 6b 2d 66 69 70 73 20 50 40 /1.0.2k -flips PH
00a0 50 2f 37 2e 34 2e 33 33 20 6d 6f 64 5f 70 65 72 P/7.4.33 mod_per
00b0 6c 2f 32 2e 30 2e 31 31 20 50 65 72 6c 2f 76 35 /2.0.11 Perl/v5
00c0 2e 31 36 2e 33 0d 0a 4c 61 73 74 2d 4d 6f 64 69 ,16.3 -L ast-Modi
00d0 66 69 65 64 3a 20 53 61 74 2c 20 36 20 46 65 fied: Sa t, 06 Fe
00e0 62 20 32 30 32 31 20 31 38 3a 32 33 3a 34 37 20 b 2021 1 8:23:47
00f0 47 4d 54 0d 0a 45 54 61 67 3a 20 22 31 61 32 2d GMT: ETa g: "1a2-
0100 35 62 61 61 66 30 39 61 62 64 37 61 39 22 0d 0a 5baaf09a bd7a9"
0110 41 63 65 70 74 2d 52 61 6e 67 65 73 3a 20 62 Accept-R anges: b
0120 79 74 65 73 0d 0a 43 6f 6e 74 65 6e 74 20 4c 65 ytes -Co ntent-Le
0130 6e 67 74 68 3a 20 34 31 38 0d 0a 4b 65 65 70 2d ngth: 41 8 -Keep-
0140 41 6c 69 76 65 3a 20 74 69 6d 65 6f 75 74 3d 35 Alive: t imeout=5
0150 2c 20 6d 61 70 3d 31 30 30 0d 0a 43 6f 6e 6e 65 , max=10 0 -Conne
0160 63 74 69 6f 6e 3a 20 4b 65 65 70 2d 41 6c 69 76 ction: K eep-Aliv
0170 65 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a e-Conte nt-Type:
0180 20 74 65 78 74 2f 68 74 6d 6c 3b 20 63 68 61 72 text/ht ml; char
0190 73 65 74 3d 55 54 46 2d 30 0d 0a 0d 0a 3c 54 49 set=UTF- 8 -<TI
01a0 54 4c 45 3e 55 70 6c 6f 61 64 20 70 61 67 65 20 TLE=Uplo ad page
01b0 66 6f 72 20 54 43 50 20 57 69 72 65 73 68 61 72 for TCP Wireshar
01c0 6b 20 4c 61 62 3c 2f 54 49 54 4c 45 3e 0a 3c 62 k Lab</T TLE>-<b
01d0 6f 64 79 20 62 67 63 6f 6e 6f 72 3d 22 23 46 46 ody bgco lor="#FF
01e0 46 46 46 46 22 3e 0a 3c 70 3e 3c 66 6f 6e 74 20 FFFF">< p<font
01f0 66 61 63 65 3d 22 41 72 69 61 6c 2c 20 48 65 6c face="Ar ial, Hel
0200 76 65 74 69 63 61 2c 20 73 61 6e 73 2d 73 65 72 vetica, sans-ser
0210 69 66 22 20 73 69 7a 65 3d 22 34 22 3e 20 43 6f if" size ="14"> Go
0220 6e 67 72 61 74 75 6c 61 74 69 6f 6e 73 21 20 3c ngratula tions! <
0230 62 72 3e 20 3c 2f 66 6f 6e 74 3e 0a 3c 50 3e br> </fo nt>-<p>
0240 3c 66 6f 6e 74 20 66 61 63 65 3d 22 41 72 69 61 <font fa ce="Aria
0250 6c 2c 20 48 65 6c 76 65 74 69 63 61 2c 20 73 61 l, Helve tica, sa
0260 6e 73 2d 73 65 72 69 66 22 3e 20 59 6f 75 27 76 ns-serif "> You'v
0270 65 20 6e 6f 77 20 74 72 61 6e 73 66 65 72 72 65 e now tr ansferre

- The absolute sequence number of the TCP SYN segment is 399438804 and the relative sequence number required is 0; The SYN flag is set to 1 in the Flags Section showing that it is a SYN segment.

Wireshark packet capture showing a TCP SYN segment. The packet list shows a SYN segment from 172.24.3.164 to 128.119.245.12. The packet details show the SYN flag is set, and the sequence number is 399438804 (raw) and 0 (relative). The packet bytes show the raw data of the SYN segment.

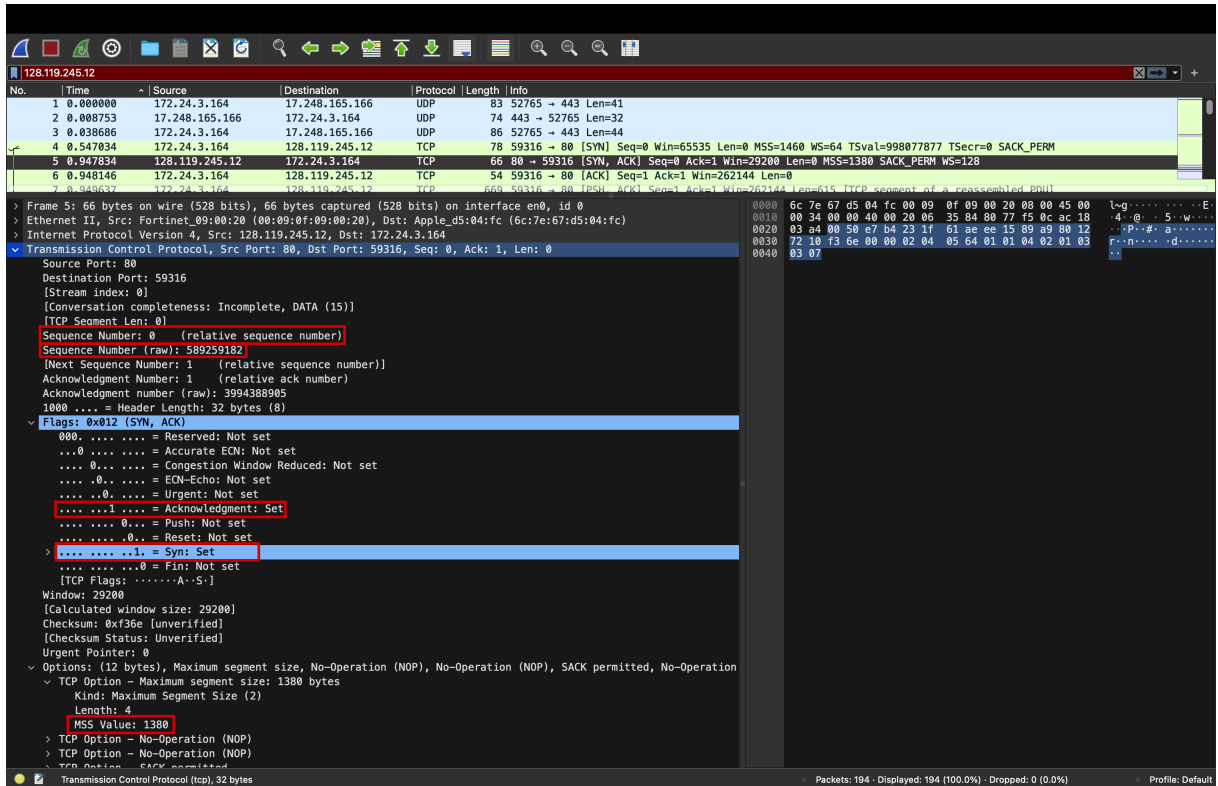
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	172.24.3.164	17.248.165.166	UDP	83	52765 → 443 Len=41
2	0.008753	17.248.165.166	172.24.3.164	UDP	74	443 → 52765 Len=32
3	0.036866	172.24.3.164	17.248.165.166	UDP	86	52765 → 443 Len=44
4	0.547034	172.24.3.164	128.119.245.12	TCP	78	59316 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=998077877 TSecr=0 SACK_PERM
5	0.947834	128.119.245.12	172.24.3.164	TCP	66	80 → 59316 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1380 SACK_PERM WS=128
6	0.948146	172.24.3.164	128.119.245.12	TCP	54	59316 → 80 [ACK] Seq=1 Ack=1 Win=262144 Len=0
7	0.949637	172.24.3.164	128.119.245.12	TCP	669	59316 → 80 [PSH, ACK] Seq=1 Ack=1 Win=262144 Len=615 [TCP segment of a reassembled PDU]
8	0.950209	172.24.3.164	128.119.245.12	TCP	191	59316 → 80 [PSH, ACK] Seq=616 Ack=1 Win=262144 Len=137 [TCP segment of a reassembled PDU]
9	0.950818	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=753 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
10	0.950843	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=2133 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
11	0.950857	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=3513 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
12	0.950873	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=4893 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
13	0.950889	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=6273 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
14	0.950903	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=7653 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
15	0.950918	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=9033 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
16	0.950932	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=10413 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
17	0.950948	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=11793 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
18	1.106171	172.24.3.164	152.195.38.76	TCP	66	59313 → 80 [FIN, ACK] Seq=1 Ack=1 Win=43 Len=0 TSval=407064428 TSecr=3329761516

Frame 4: 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface en0, id 0
 Ethernet II, Src: Apple_d5:84:fc (6c:7e:67:d5:04:fc), Dst: Fortinet_09:00:20 (00:09:0f:09:00:20)
 Internet Protocol Version 4, Src: 172.24.3.164, Dst: 128.119.245.12
 Transmission Control Protocol, Src Port: 59316, Dst Port: 80, Seq: 0, Len: 0
 Source Port: 59316
 Destination Port: 80
 [Stream index: 0]
 [Conversation completeness: Incomplete, DATA (15)]
 [TCP Segment Len: 0]
 Sequence Number: 0 (relative sequence number)
 Sequence Number (raw): 399438804
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 0
 Acknowledgment number (raw): 0
 1011... = Header Length: 44 bytes (11)
 Flags: 0x002 (SYN)
 000. = Reserved: Not set
 ...0. = Accurate ECN: Not set
 0... = Congestion Window Reduced: Not set
0.. = ECN-Echo: Not set
0. = Urgent: Not set
0. = Acknowledgment: Not set
0.. = Push: Not set
0.. = Reset: Not set
 >1. = Syn: Set
0. = Fin: Not set
 [TCP Flags:S-]
 Window: 65535

Syn (tcp.flags.syn), 1 byte

Packets: 194 · Displayed: 194 (100.0%) · Dropped: 0 (0.0%) · Profile: Default

4. The absolute sequence number of the TCP SYN segment is 589259182 and the relative sequence number required is 0; The value of Acknowledgement number field is 1; The server adds 1 to the sequence number of the received SYN segment; The SYNACK segment is identified as both SYN flag and Acknowledgement in the segment are set to 1. The value of MSS indicated in the Options field of the SYNACK segment is 1380 bytes.



5. Note that numbers in frames 7,8,9,10,11,12 is the TCP segments carrying the first six partitions of the HTTP POST message, and numbers 21,22,23 are the corresponding ACK to the segments in frame mentioned repectively.

	Seq. no.	Sent time	ACK seg. no.	ACK recieved time
Segment 1	1	21	0.949637	1.195193
Segment 2	616	22	0.950209	1.195196
Segment 3	753	23	0.950818	1.195196
Segment 4	2133	23	0.950843	1.195196
Segment 5	3513	23	0.950857	1.195196
Segment 6	4893	23	0.950873	1.195196

Since we have the typical value of $\alpha = 0.125$.

$$\begin{aligned}
EstimatedRTT &= (1 - \alpha) \times EstimatedRTT + \alpha \times SampleRTT \\
&= (1 - 0.125) \times EstimatedRTT + 0.125 \times SampleRTT \\
&= 0.875 \times EstimatedRTT + 0.125 \times SampleRTT
\end{aligned}$$

	RTT	EstimatedRTT
Segment 1	$1.195193 - 0.949637 = 0.245556$	0.245556
Segment 2	$1.195196 - 0.950209 = 0.244987$	$0.875 \times 0.245556 + 0.125 \times 0.244987 = 0.245485$
Segment 3	$1.195196 - 0.950818 = 0.244378$	$0.875 \times 0.245485 + 0.125 \times 0.244378 = 0.245347$
Segment 4	$1.195196 - 0.950843 = 0.244353$	$0.875 \times 0.245347 + 0.125 \times 0.244353 = 0.245223$
Segment 5	$1.195196 - 0.950857 = 0.244339$	$0.875 \times 0.245223 + 0.125 \times 0.244339 = 0.245113$
Segment 6	$1.195196 - 0.950873 = 0.244323$	$0.875 \times 0.245113 + 0.125 \times 0.244323 = 0.245014$

Wireshark packet capture analysis showing a list of network packets. The selected packet (No. 7) is a TCP segment from 172.24.3.164 to 128.119.245.12, Seq=669, Ack=1, Win=262144, Len=615. The packet details pane shows the Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol (TCP) layers. The packet bytes pane displays the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	172.24.3.164	17.248.165.166	UDP	83	52765 → 443 Len=41
2	0.008753	172.24.3.164	172.24.3.164	UDP	74	443 → 52765 Len=32
3	0.036896	172.24.3.164	17.248.165.166	UDP	86	52765 → 443 Len=44
4	0.507834	172.24.3.164	128.119.245.12	TCP	78	59316 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=998077877 TSecr=0 SACK_PERM
5	0.947834	128.119.245.12	172.24.3.164	TCP	66	80 → 59316 [SYN, ACK] Seq=0 Ack=1 Win=29208 Len=0 MSS=1380 SACK_PERM WS=128
6	0.948146	172.24.3.164	128.119.245.12	TCP	54	59316 → 80 [ACK] Seq=1 Ack=1 Win=262144 Len=0
7	0.949637	172.24.3.164	128.119.245.12	TCP	669	59316 → 80 [PSH, ACK] Seq=1 Ack=1 Win=262144 Len=615 [TCP segment of a reassembled PDU]
8	0.950209	172.24.3.164	128.119.245.12	TCP	191	59316 → 80 [PSH, ACK] Seq=616 Ack=1 Win=262144 Len=137 [TCP segment of a reassembled PDU]
9	0.950818	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=753 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
10	0.950843	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=2133 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
11	0.950857	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=3513 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
12	0.950873	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=4893 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
13	0.950889	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=6273 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
14	0.950993	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=7653 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
15	0.950918	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=9033 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
16	0.950932	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=10413 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
17	0.950948	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=11793 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
18	1.106171	172.24.3.164	152.195.38.76	TCP	66	59313 → 80 [FIN, ACK] Seq=1 Ack=1 Win=43 Len=0 TSval=407064428 TSecr=3329761516
19	1.110793	152.195.38.76	172.24.3.164	TCP	66	80 → 59313 [FIN, ACK] Seq=1 Ack=2 Win=131 Len=0 TSval=3329791584 TSecr=407064428
20	1.111052	152.195.38.76	172.24.3.164	TCP	66	59313 → 80 [ACK] Seq=2 Ack=2 Win=43 Len=0 TSval=407064433 TSecr=3329791584
21	1.195193	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=616 Win=30464 Len=0
22	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=753 Win=31744 Len=0
23	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=6273 Win=42752 Len=0
24	1.195197	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=13173 Win=56576 Len=0
25	1.195468	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=13173 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
26	1.195644	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=14553 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
27	1.195674	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=15933 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
28	1.195689	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=17313 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
29	1.195784	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=18693 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
30	1.195723	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=20073 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
31	1.195736	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=21453 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]

Frame 7: 669 bytes on wire (5352 bits), 669 bytes captured (5352 bits) on interface en0, id 0
 Ethernet II, Src: Apple_d5:04:fc (6c:7e:67:d5:04:fc), Dst: Fortinet_09:00:20 (00:09:0f:09:00:20)
 Internet Protocol Version 4, Src: 172.24.3.164, Dst: 128.119.245.12
 Transmission Control Protocol, Src Port: 59316, Dst Port: 80, Seq: 1, Ack: 1, Len: 615

Bytes 64-668: TCP segment data (tcp.segment_data) Packets: 194 - Displayed: 194 (100.0%) Profile: Default

Wireshark packet capture analysis showing a list of network packets. The selected packet (No. 21) is a TCP segment from 128.119.245.12 to 172.24.3.164, Seq=56, Ack=616, Win=30464, Len=0. The packet details pane shows the Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol (TCP) layers. The packet bytes pane displays the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
15	0.950918	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=9033 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
16	0.950932	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=10413 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
17	0.950948	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=11793 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
18	1.106171	172.24.3.164	152.195.38.76	TCP	66	59313 → 80 [FIN, ACK] Seq=1 Ack=1 Win=43 Len=0 TSval=407064428 TSecr=3329761516
19	1.110793	152.195.38.76	172.24.3.164	TCP	66	80 → 59313 [FIN, ACK] Seq=1 Ack=2 Win=131 Len=0 TSval=3329791584 TSecr=407064428
20	1.111052	152.195.38.76	172.24.3.164	TCP	66	59313 → 80 [ACK] Seq=2 Ack=2 Win=43 Len=0 TSval=407064433 TSecr=3329791584
21	1.195193	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=616 Win=30464 Len=0
22	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=753 Win=31744 Len=0
23	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=6273 Win=42752 Len=0
24	1.195197	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=13173 Win=56576 Len=0
25	1.195468	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=13173 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
26	1.195644	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=14553 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
27	1.195674	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=15933 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
28	1.195689	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=17313 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
29	1.195784	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=18693 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
30	1.195723	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=20073 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
31	1.195736	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=21453 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]

Frame 21: 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface en0, id 0
 Ethernet II, Src: Fortinet_09:00:20 (00:09:0f:09:00:20), Dst: Apple_d5:04:fc (6c:7e:67:d5:04:fc)
 Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.24.3.164
 Transmission Control Protocol, Src Port: 80, Dst Port: 59316, Seq: 1, Ack: 616, Len: 0

Source Port: 80
 Destination Port: 59316
 [Stream index: 0]
 [Conversation completeness: Incomplete, DATA (15)]
 [TCP Segment Len: 0]
 Sequence Number: 1 (relative sequence number)
 Sequence Number (raw): 589259183
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 616 (relative ack number)
 Acknowledgment number (raw): 3994389520
 0101 = Header Length: 20 bytes (5)
 > Flags: 0x010 (ACK)
 Window: 238
 [Calculated window size: 30464]
 [Window size scaling factor: 128]
 Checksum: 0xa2ac [unverified]
 [Checksum Status: Unverified]
 Urgent Pointer: 0
 > [Timestamps]
 > [SEQ/ACK analysis]
 [This is an ACK to the segment in frame 7]
 [The RTT to ACK the segment was: 0.245550000 seconds]
 [RTT: 0.481112000 seconds]

Transmission Control Protocol (tcp), 20 bytes Packets: 194 - Displayed: 194 (100.0%) - Dropped: 0 (0.0%) Profile: Default

Wireshark packet capture analysis showing a TCP segment (Seq=753, Win=31744) and its acknowledgment (Seq=6273, Win=42752). The packet list shows a sequence of TCP segments from 15 to 31. The packet details pane shows the structure of the TCP segment, including the source port (80), destination port (59316), sequence number (753), and acknowledgment number (6273). The packet bytes pane shows the raw data of the segment.

No.	Time	Source	Destination	Protocol	Length	Info
15	0.950918	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=9033 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
16	0.950932	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=10413 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
17	0.950948	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=11793 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
18	1.106171	172.24.3.164	152.195.38.76	TCP	66	59313 → 80 [FIN, ACK] Seq=1 Ack=1 Win=43 Len=0 TSval=407064428 TSecr=3329761516
19	1.110793	152.195.38.76	172.24.3.164	TCP	66	80 → 59313 [FIN, ACK] Seq=1 Ack=2 Win=131 Len=0 TSval=3329791584 TSecr=407064428
20	1.111052	172.24.3.164	152.195.38.76	TCP	66	59313 → 80 [ACK] Seq=2 Ack=2 Win=43 Len=0 TSval=407064433 TSecr=3329791584
21	1.195193	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=616 Win=30464 Len=0
22	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=753 Win=31744 Len=0
23	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=6273 Win=42752 Len=0
24	1.195197	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=13173 Win=56576 Len=0
25	1.195468	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=13173 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
26	1.195644	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=14553 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
27	1.195674	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=15933 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
28	1.195689	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=17313 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
29	1.195704	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=18693 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
30	1.195723	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=20073 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
31	1.195736	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=21453 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]

Frame 22: 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface en0, id 0
 Ethernet II, Src: Fortinet_09:00:00:00:00:20 (00:09:00:00:00:20), Dst: Apple_d5:04:fc (6c:7e:67:d5:04:fc)
 Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.24.3.164
 Transmission Control Protocol, Src Port: 80, Dst Port: 59316, Seq: 1, Ack: 753, Len: 0

Source Port: 80
 Destination Port: 59316
 [Stream index: 0]
 [Conversation completeness: Incomplete, DATA (15)]
 [TCP Segment Len: 0]
 Sequence Number: 1 (relative sequence number)
 Sequence Number (raw): 589259183
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 753 (relative ack number)
 Acknowledgment number (raw): 3994389657
 0101 = Header Length: 20 bytes (5)
 > Flags: 0x010 (ACK)
 Window: 248
 [Calculated window size: 31744]
 [Window size scaling factor: 128]
 Checksum: 0xa219 [unverified]
 [Checksum Status: Unverified]
 Urgent Pointer: 0
 > [Timestamps]
 > [SEQ/ACK analysis]
 [This is an ACK to the segment in frame: 0]
 [The RTT to ACK the segment was: 0.244987000 seconds]
 [iRTT: 0.401112000 seconds]

Transmission Control Protocol (tcp), 20 bytes

Packets: 194 - Displayed: 194 (100.0%) - Dropped: 0 (0.0%)

Wireshark packet capture analysis showing a TCP segment (Seq=6273, Win=42752) and its acknowledgment (Seq=6273, Win=42752). The packet list shows a sequence of TCP segments from 15 to 31. The packet details pane shows the structure of the TCP segment, including the source port (80), destination port (59316), sequence number (6273), and acknowledgment number (6273). The packet bytes pane shows the raw data of the segment.

No.	Time	Source	Destination	Protocol	Length	Info
15	0.950918	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=9033 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
16	0.950932	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=10413 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
17	0.950948	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=11793 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
18	1.106171	172.24.3.164	152.195.38.76	TCP	66	59313 → 80 [FIN, ACK] Seq=1 Ack=1 Win=43 Len=0 TSval=407064428 TSecr=3329761516
19	1.110793	152.195.38.76	172.24.3.164	TCP	66	80 → 59313 [FIN, ACK] Seq=1 Ack=2 Win=131 Len=0 TSval=3329791584 TSecr=407064428
20	1.111052	172.24.3.164	152.195.38.76	TCP	66	59313 → 80 [ACK] Seq=2 Ack=2 Win=43 Len=0 TSval=407064433 TSecr=3329791584
21	1.195193	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=616 Win=30464 Len=0
22	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=753 Win=31744 Len=0
23	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=6273 Win=42752 Len=0
24	1.195197	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=13173 Win=56576 Len=0
25	1.195468	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=13173 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
26	1.195644	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=14553 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
27	1.195674	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=15933 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
28	1.195689	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=17313 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
29	1.195704	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=18693 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
30	1.195723	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=20073 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
31	1.195736	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=21453 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]

Frame 23: 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface en0, id 0
 Ethernet II, Src: Fortinet_09:00:00:00:00:20 (00:09:00:00:00:20), Dst: Apple_d5:04:fc (6c:7e:67:d5:04:fc)
 Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.24.3.164
 Transmission Control Protocol, Src Port: 80, Dst Port: 59316, Seq: 1, Ack: 6273, Len: 0

Source Port: 80
 Destination Port: 59316
 [Stream index: 0]
 [Conversation completeness: Incomplete, DATA (15)]
 [TCP Segment Len: 0]
 Sequence Number: 1 (relative sequence number)
 Sequence Number (raw): 589259183
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 6273 (relative ack number)
 Acknowledgment number (raw): 3994395177
 0101 = Header Length: 20 bytes (5)
 > Flags: 0x010 (ACK)
 Window: 334
 [Calculated window size: 42752]
 [Window size scaling factor: 128]
 Checksum: 0x8c33 [unverified]
 [Checksum Status: Unverified]
 Urgent Pointer: 0
 > [Timestamps]
 > [SEQ/ACK analysis]
 [This is an ACK to the segment in frame: 12]
 [The RTT to ACK the segment was: 0.244323000 seconds]
 [iRTT: 0.401112000 seconds]

Transmission Control Protocol (tcp), 20 bytes

Packets: 194 - Displayed: 194 (100.0%) - Dropped: 0 (0.0%)

- As seen from the Len field in the Info column, the size of the payload in the first TCP segment is 615 bytes, the second is 137 bytes, and the rest are 1380 bytes.

The image shows a Wireshark packet capture analysis. The top pane displays a list of packets. The bottom pane shows the details of a selected packet (Frame 7).

Packets List:

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	172.24.3.164	17.248.165.166	UDP	83	52765 → 443 Len=41
2	0.000753	17.248.165.166	172.24.3.164	UDP	74	443 → 52765 Len=32
3	0.038686	172.24.3.164	17.248.165.166	UDP	86	52765 → 443 Len=44
4	0.547834	172.24.3.164	128.119.245.12	TCP	78	59316 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=998077877 TSecr=0 SACK_PERM
5	0.947834	128.119.245.12	172.24.3.164	TCP	66	80 → 59316 [SYN, ACK] Seq=0 Ack=1 Win=29280 Len=0 MSS=1380 SACK_PERM WS=128
6	0.948146	172.24.3.164	128.119.245.12	TCP	54	59316 → 80 [ACK] Seq=1 Ack=1 Win=262144 Len=0
7	0.949637	172.24.3.164	128.119.245.12	TCP	669	59316 → 80 [PSH, ACK] Seq=1 Ack=1 Win=262144 Len=615 [TCP segment of a reassembled PDU]
8	0.950209	172.24.3.164	128.119.245.12	TCP	191	59316 → 80 [PSH, ACK] Seq=616 Ack=1 Win=262144 Len=137 [TCP segment of a reassembled PDU]
9	0.950818	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=753 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
10	0.950843	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=2133 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
11	0.950857	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=3513 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
12	0.950873	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=4893 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
13	0.950889	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=6273 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
14	0.950903	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=7653 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
15	0.950918	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=9833 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
16	0.950932	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=10413 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
17	0.950948	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=11793 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
18	1.106171	172.24.3.164	152.105.38.76	TCP	66	59313 → 80 [FIN, ACK] Seq=1 Ack=1 Win=43 Len=0 TSval=487064428 TSecr=3329761516

Frame 7: 669 bytes on wire (5352 bits), 669 bytes captured (5352 bits) on interface en0, id 0

Ethernet II, Src: Apple_d5:04:fc (6c:7e:67:d5:04:fc), Dst: Fortinet_09:00:20 (00:09:0f:09:00:20)

Internet Protocol Version 4, Src: 172.24.3.164, Dst: 128.119.245.12

Transmission Control Protocol, Src Port: 59316, Dst Port: 80, Seq: 1, Ack: 1, Len: 615

Source Port: 59316

Destination Port: 80

[Stream index: 0]

[Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 615]

Sequence Number: 1 (relative sequence number)

Sequence Number (raw): 3994388905

[Next Sequence Number: 616 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

Acknowledgment number (raw): 589259183

0101 = Header Length: 20 bytes (5)

Flags: 0x018 (PSH, ACK)

Window: 4896

[Calculated window size: 262144]

[Window size scaling factor: 64]

Checksum: 0xf60c (Unverified)

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

[SEQ/ACK analysis]

TCP payload (615 bytes)

Reassembled PDU in frame: 147

TCP segment data (615 bytes)

0020 f5 0c e7 b4 00 00 ee 15 89 a9 23 1f 61 af 50 18P...a:P
0030 10 00 f6 0c 00 00 50 4f 53 54 20 2f 77 69 72 65PO ST /wire
0040 73 68 61 72 6b 2d 6c 61 62 73 2f 6c 61 62 33 2d shark-la bs/lab3-
0050 31 2d 72 65 70 6c 79 2e 68 74 6d 20 48 54 54 50 1-reply, htm HTTP
0060 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 67 61 69 61 /1.1: Ho st: gaia
0070 2e 63 73 2e 75 6d 61 73 73 2e 65 64 75 0d 0a 4f .cs.umas s.edu:0
0080 72 69 67 69 66 3a 20 68 74 74 70 3a 2f 2f 67 61 rigin: h ttp://ga
0090 69 61 2e 63 73 2e 75 6d 61 73 73 2e 65 64 75 6d ia.cs.um ass.edu
00a0 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20 6d .Content -Type: m
00b0 75 6c 74 69 70 61 72 74 2f 66 6f 72 6d 2d 64 61 ultipart /form-da
00c0 74 61 3b 20 62 6f 75 6e 64 61 72 79 3d 2d 2d 2d ta; bou n dary=--
00d0 2d 57 65 62 4b 69 74 46 6f 72 6d 42 6f 75 6e 64 -WebXtlf ormsound
00e0 61 72 79 45 53 48 56 6c 31 74 51 72 31 72 43 68 aryESHLV itQrlrCh
00f0 77 74 4c 0d 0a 41 63 63 65 70 74 2d 45 6e 63 6f wtl- Acc ep t-Enco
0100 64 69 6e 67 3a 20 67 7a 69 70 2c 20 64 65 66 6c ding: gz ip, defl
0110 61 74 65 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a ate: Con nection:
0120 20 6b 65 65 70 2d 61 6c 69 76 65 0d 0a 55 70 67 keep-al ive: Upg
0130 72 61 64 65 2d 49 6e 73 65 63 75 72 65 2d 52 65 rade-Ins ecure-Re
0140 71 75 65 73 74 73 3a 20 31 0d 0a 41 63 63 65 70 quests: 1 'Accep
0150 74 3a 20 74 65 70 74 2f 68 74 6d 6c 2c 61 70 70 t: text/ html, app
0160 6c 69 63 61 74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 Location /xhtml/x
0170 6d 6c 2c 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 m l, appli cation/x
0180 6d 6c 3b 71 3d 30 2e 39 2c 2a 2f 2a 3b 71 3d 30 m l; q=0.9 ,*/; q=0
0190 2e 38 0d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 .8 'User -Agent:
01a0 4d 6f 7a 69 6c 6c 61 2f 35 2e 30 20 28 4d 61 63 Mozilla/ 5.0 (Mac
01b0 69 6e 74 6f 73 68 3b 20 49 6e 74 65 6c 20 4d 61 intosh; Intel Ma
01c0 63 20 4f 53 20 58 20 31 30 5f 31 35 5f 37 29 20 c OS X 1 0.15.7)
01d0 41 70 70 6c 65 57 65 62 4b 69 74 2f 3b 30 35 2e AppleWeb KIt/605
01e0 31 2e 31 35 20 28 4b 48 54 4d 4c 2c 20 6c 69 6b 1-15 (KH TML, lik
01f0 65 20 47 65 63 6b 6f 29 20 56 65 72 73 69 6f 6e e Gecko) Version
0200 2f 31 36 2e 36 20 53 61 66 61 72 69 2f 36 30 35 /16.6 Sa fari/605
0210 2e 31 2e 31 35 0d 0a 52 65 65 65 72 65 72 3a 20 .1.15 R eferer:
0220 68 74 70 3a 2f 2f 67 61 69 61 2e 63 73 2e 75 http://g aia.cs.u

Packets: 194 - Displayed: 194 (100.0%) - Dropped: 0 (0.0%) Profile: Default

- As seen from the Win field in the Info column, the minimum size of available receiver buffer space advertised by the server in the entire trace is 29200; and the maximum is 181632.

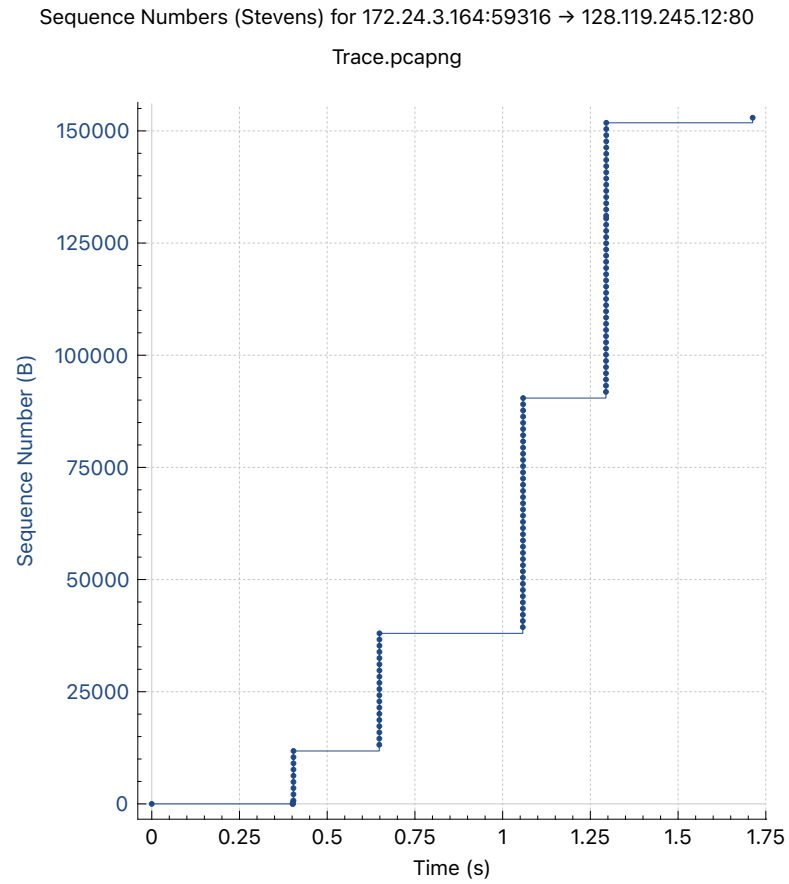
The image shows a Wireshark network traffic capture. The top pane displays a list of 158 packets. The bottom pane shows the details of the selected packet (No. 5, Time 0.947834, Source 128.119.245.12, Destination 172.24.3.164, Protocol TCP, Length 66, Info 80 → 59316 [ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1380 SACK_PERM WS=128).

No.	Time	Source	Destination	Protocol	Length	Info
5	0.947834	128.119.245.12	172.24.3.164	TCP	66	80 → 59316 [ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1380 SACK_PERM WS=128
21	1.195193	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=616 Win=30464 Len=0
22	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=753 Win=31744 Len=0
23	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=6273 Win=42752 Len=0
24	1.195197	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=13173 Win=56576 Len=0
44	1.604159	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=20073 Win=70400 Len=0
45	1.604161	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=26973 Win=84224 Len=0
46	1.604161	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=33873 Win=97920 Len=0
47	1.604162	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=39393 Win=109056 Len=0
95	1.841174	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=46293 Win=122752 Len=0
96	1.841176	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=53193 Win=136576 Len=0
97	1.841176	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=60093 Win=150400 Len=0
98	1.841177	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=66993 Win=164224 Len=0
99	1.841177	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=73893 Win=178048 Len=0
100	1.841178	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=80793 Win=179584 Len=0
101	1.841178	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=87693 Win=179584 Len=0
102	1.841178	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=94593 Win=179584 Len=0
148	2.259270	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=98733 Win=179584 Len=0
149	2.259272	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=105633 Win=174464 Len=0
150	2.259272	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=112533 Win=179584 Len=0
151	2.259272	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=119433 Win=179584 Len=0
152	2.259273	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=126333 Win=179584 Len=0
153	2.259273	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=133233 Win=180608 Len=0
154	2.259273	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=140133 Win=179584 Len=0
155	2.259274	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=147033 Win=179584 Len=0
156	2.259274	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=153933 Win=179584 Len=0
157	2.259274	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=160833 Win=179584 Len=0
158	2.259274	128.119.245.12	172.24.3.164	HTTP	831	HTTP/1.1 200 OK (text/html)

Frame 5: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface en0, id 0
 Ethernet II, Src: Fortinet_09:00:20 (00:09:0f:09:00:20), Dst: Apple_d5:04:fc (6c:7e:67:d5:04:fc)
 Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.24.3.164
 Transmission Control Protocol, Src Port: 80, Dst Port: 59316, Seq: 0, Ack: 1, Len: 0
 Source Port: 80
 Destination Port: 59316
 [Stream index: 0]
 [Conversation completeness: Incomplete, DATA (15)]
 [TCP Segment Len: 0]
 Sequence Number: 0 (relative sequence number)
 Sequence Number (raw): 589259182
 [Next Sequence Number: 1 (relative sequence number)]
 Acknowledgment Number: 1 (relative ack number)
 Acknowledgment number (raw): 3994388905
 1000 = Header Length: 32 bytes (8)

Packets: 194 · Displayed: 28 (14.4%) · Dropped: 0 (0.0%) · Profile: Default

8. There is no retransmitted segments in the trace since there is no sequence number smaller than the previous segments. Notice that from the Time-Sequence-Graph (Stevens), the sequence numbers of the TCP segments from the the source to the destination increase monotonically across time.



9. The difference between the acknowledged sequence numbers of two consecutive ACKs indicates the data received by the server between these two ACKs.

No.	Time	Source	Destination	Protocol	Length	Info
5	0.947834	128.119.245.12	172.24.3.164	TCP	66	80 → 59316 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1380 SACK_PERM WS=128
21	1.195193	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=616 Win=30464 Len=0
22	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=753 Win=31744 Len=0
23	1.195196	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=6273 Win=42752 Len=0
24	1.195197	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=13173 Win=56576 Len=0
44	1.604159	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=20073 Win=70400 Len=0
45	1.604161	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=26973 Win=84224 Len=0
46	1.604161	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=33873 Win=97920 Len=0
47	1.604162	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=39393 Win=109056 Len=0
95	1.841174	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=46293 Win=122752 Len=0
96	1.841176	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=53193 Win=136576 Len=0
97	1.841176	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=60093 Win=150400 Len=0
98	1.841177	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=66993 Win=164224 Len=0
99	1.841177	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=73893 Win=178048 Len=0
100	1.841178	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=80793 Win=179584 Len=0
101	1.841178	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=87693 Win=179584 Len=0
102	1.841178	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=91833 Win=181632 Len=0
148	2.259270	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=98733 Win=179584 Len=0
149	2.259272	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=105633 Win=174464 Len=0
150	2.259272	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=112533 Win=179584 Len=0
151	2.259272	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=119433 Win=179584 Len=0
152	2.259273	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=126333 Win=179584 Len=0
153	2.259273	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=131101 Win=180608 Len=0
154	2.259273	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=132481 Win=179584 Len=0
155	2.259274	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=139381 Win=179584 Len=0
156	2.259274	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=146281 Win=179584 Len=0
157	2.259274	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=152937 Win=179584 Len=0
158	2.259274	128.119.245.12	172.24.3.164	HTTP	831	HTTP/1.1 200 OK (text/html)

> Ethernet II, Src: portnet_wy00:20 (00:09:0f:09:00:20), Dst: Apple_d3:04:1c (0c:e3:0d:04:1c)
 > Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.24.3.164
 > Transmission Control Protocol, Src Port: 80, Dst Port: 59316, Seq: 0, Ack: 1, Len: 0

Packets: 194 - Displayed: 28 (14.4%) - Dropped: 0 (0.0%) Profile: Default

	Acknowledged Number	Acknowledged Data
Segment 1	616	616 - 1 = 615
Segment 2	753	753 - 616 = 137
Segment 3	6273	6273 - 753 = 5520
Segment 4	13173	13173 - 6273 = 6900
Segment 5	20073	20073 - 13173 = 6900
Segment 6	26973	26973 - 20073 = 6900

10. The time when the client sends the first TCP segment carrying the HTTP POST message is 0.949637. And the time when the ACK for the last TCP segment carrying the HTTP POST message is received from the server is 2.259274.

From this, we have the total time for transmission = $2.259274 - 0.949637 = 1.309637$

The sequence number of the first TCP segment is 1 byte. The Acknowledged sequence number of the last ACK is 152937 bytes.

From this, we have the total amount of transmitted data = $152937 - 1 = 152936$ bytes.

Since the throughput can be calculated from dividing the total amount of transmitted data by the total time for transmission.

That is $\text{throughput} = 152936 / 1.309637 = 116777.397096$ bytes/sec.

Apply a display filter... <34/

Continuity
 Learn how to use your devices together.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	172.24.3.164	17.248.165.166	UDP	83	52765 → 443 Len=1
2	0.008753	17.248.165.166	172.24.3.164	UDP	74	443 → 52765 Len=32
3	0.038686	172.24.3.164	17.248.165.166	UDP	86	52765 → 443 Len=1
4	0.547034	172.24.3.164	128.119.245.12	TCP	78	59316 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=64 TSval=998077877 TSecn=0 SACK_PERM
5	0.947834	128.119.245.12	172.24.3.164	TCP	66	80 → 59316 [RST, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1380 SACK_PERM_WS=128
6	0.948146	172.24.3.164	128.119.245.12	TCP	54	59316 → 80 [ACK] Seq=1 Ack=1 Win=262144 Len=0
7	0.949637	172.24.3.164	128.119.245.12	TCP	669	59316 → 80 [PSH, ACK] Seq=1 Ack=1 Win=262144 Len=615 [TCP segment of a reassembled PDU]
8	0.950209	172.24.3.164	128.119.245.12	TCP	191	59316 → 80 [PSH, ACK] Seq=616 Ack=1 Win=262144 Len=113 [TCP segment of a reassembled PDU]
9	0.950818	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=753 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
10	0.950843	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=2133 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
11	0.950857	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=3513 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
12	0.950873	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=4893 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
13	0.950889	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=6273 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
14	0.950903	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=7653 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
15	0.950918	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=9033 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
16	0.950932	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=10413 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
17	0.950948	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=11793 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
18	1.106121	172.24.3.164	157.105.38.76	TCP	66	59313 → 80 [FIN, ACK] Seq=1 Ack=1 Win=43 Len=0 TSval=107064478 TSecn=332076516

> Frame 7: 669 bytes on wire (5352 bits), 669 bytes captured (5352 bits) on interface en0, id 0

> Ethernet II, Src: Apple_d5:04:fc (6c:7e:67:d5:04:fc), Dst: Fortinet_09:00:20 (00:09:0f:00:20:00)

> Internet Protocol Version 4, Src: 172.24.3.164, Dst: 128.119.245.12

> Transmission Control Protocol, Src Port: 59316, Dst Port: 80, Seq: 1, Ack: 1, Len: 615

Source Port: 59316

Destination Port: 80

[Stream index: 0]

[Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 615]

Sequence Number: 1 (relative sequence number)

Sequence Number (raw): 3994388905

Next Sequence Number: 616 (relative sequence number)

Acknowledgment Number: 1 (relative ack number)

Acknowledgment Number (raw): 589259183

0101 ... = Header Length: 20 bytes (5)

> Flags: 0x018 (PSH, ACK)

Window: 4096

[Calculated window size: 262144]

Window size scaling factor: 64

Checksum: 0x60c (unverified)

[Checksum Status: Unverified]

Urgent Pointer: 0

[Timestamps]

> [SEQ/ACK analysis]

[IRTT: 0.401128000 seconds]

[Bytes in Flight: 615]

[Bytes sent since last PSH flag: 615]

TCP payload (615 bytes)

[Reassembled PDU in frames: 147]

The scaled window size (if scaling has been used) (tcp.window_size), 2 bytes

0030 10 00 f6 c0 00 80 50 4f 53 54 20 2f 7f 69 72 65 ...=PO ST /wire

0031 73 68 61 72 60 20 6c 61 62 73 74 6c 61 62 33 20 ...shark-ls bs/lab3

0032 00 31 20 72 65 70 6c 75 2e 68 74 6d 20 68 54 50 ...l-copy, htm HTTP

0033 00 2f 31 21 31 0d 0a 4b 73 74 3a 20 67 61 69 61 .../1.1. Ho st: gaia

0034 00 70 26 32 7e 25 6d 61 63 25 65 64 75 0d 04 6f ...cs.umass.edu. 0

0035 00 74 70 67 69 6e 3a 20 68 74 70 3a 21 2f 61 61 ...rgins: h ttps://o

0036 00 69 61 2e 63 73 25 7e 6d 61 73 73 2e 65 64 70 ...iacs.um ass.edu

0037 00 0a 43 6f 6e 74 65 64 74 2d 54 79 70 65 3a 20 6d ...Content -type: m

0038 00 75 6c 74 69 70 61 72 74 2f 26 6f 67 62 6d 20 64 61 ...upload /form-da

0039 00 74 61 3b 20 62 75 6e 64 61 63 74 69 65 64 61 63 ...tag, boum dary=

0040 00 29 57 65 62 6b 69 74 46 6f 72 6d 62 6d 62 65 64 ...-Webkit ormbound

0041 00 61 72 79 45 53 48 56 6c 31 74 51 72 31 72 63 68 6f ...arvESHV1 l0r10rkh

0042 00 77 74 4c 0d 61 63 63 65 70

Wireshark packet capture analysis showing a list of network packets and a detailed view of a selected packet (No. 157).

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
145	1.842330	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=149041 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
146	1.842343	172.24.3.164	128.119.245.12	TCP	1434	59316 → 80 [ACK] Seq=150421 Ack=1 Win=262144 Len=1380 [TCP segment of a reassembled PDU]
147	1.842363	172.24.3.164	128.119.245.12	HTTP	1190	POST /wirespark-labs/lab3-1-reply.htm HTTP/1.1 (text/plain)
148	2.259270	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=98733 Win=179584 Len=0
149	2.259272	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=105633 Win=174464 Len=0
150	2.259272	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=112533 Win=179584 Len=0
151	2.259272	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=119433 Win=179584 Len=0
152	2.259273	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=126333 Win=179584 Len=0
153	2.259273	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=131101 Win=180608 Len=0
154	2.259273	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=132481 Win=179584 Len=0
155	2.259274	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=139381 Win=179584 Len=0
156	2.259274	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=146281 Win=179584 Len=0
157	2.259274	128.119.245.12	172.24.3.164	TCP	56	80 → 59316 [ACK] Seq=1 Ack=152937 Win=179584 Len=0
158	2.259274	128.119.245.12	172.24.3.164	HTTP	831	HTTP/1.1 200 OK (text/html)
159	2.259581	172.24.3.164	128.119.245.12	TCP	54	59316 → 80 [ACK] Seq=152937 Ack=778 Win=261312 Len=0
160	2.259674	172.24.3.164	35.190.88.7	TCP	54	59304 → 443 [ACK] Seq=1 Ack=1 Win=2048 Len=0
161	2.259726	35.190.88.7	172.24.3.164	TCP	66	[TCP ACKed unseen segment] 443 → 59304 [ACK] Seq=1 Ack=2 Win=265 Len=0 TSval=505485294 TSecr=1009552702

Packet Details (No. 157):

- Ethernet II, Src: Fortinet_09:00:20 (00:09:0f:00:00:20), Dst: Apple_d5:04:fc (6c:7e:67:d5:04:fc)
- Internet Protocol Version 4, Src: 128.119.245.12, Dst: 172.24.3.164
- Transmission Control Protocol, Src Port: 80, Dst Port: 59316, Seq: 1, Ack: 152937, Len: 0
 - Source Port: 80
 - Destination Port: 59316
 - [Stream index: 0]
 - [Conversation completeness: Incomplete, DATA (15)]
 - [TCP Segment Len: 0]
 - Sequence Number: 1 (relative sequence number)
 - Sequence Number (raw): 589259183
 - [Next Sequence Number: 1 (relative sequence number)]
 - Acknowledgment Number: 152937 (relative ack number)
 - Acknowledgment number (raw): 3994541841
 - 0101 = Header Length: 20 bytes (5)
 - Flags: 0x010 (ACK)
 - Window: 1403
 - [Calculated window size: 179584]
 - [Window size scaling factor: 128]
 - Checksum: 0x4b1c (unverified)
 - [Checksum Status: Unverified]
 - Urgent Pointer: 0
- [Timestamps]
 - [Time since first frame in this TCP stream: 1.712240000 seconds]
 - [Time since previous frame in this TCP stream: 0.000000000 seconds]
- [SEQ/ACK analysis]
 - [This is an ACK to the segment in frame: 147]
 - [The RTT to ACK the segment was: 0.416911000 seconds]
 - [JRTT: 0.401112000 seconds]

Trace.pcapng

Packets: 194 - Displayed: 194 (100.0%) - Dropped: 0 (0.0%)

Profile: Default